

INTELLIGENT DATA CLEANING & VALIDATION BOT

Enterprise Deployment Overview

Version 1.0

1. Executive Summary

Organizations routinely handle Excel / CSV datasets that are incomplete, inconsistent, and error-prone. Manual cleaning is slow, non-scalable, and audit-invisible.

The Intelligent Data Cleaning & Validation Bot is an enterprise-grade, containerized system that:

- Standardizes Excel datasets using deterministic cleaning logic
- Generates a structured audit log for flagged and rejected data
- Prevents duplicate file processing while allowing controlled re-processing after a job has been completed. Re-processing is allowed; however, all attempts to process same file are counted and logged.
- Supports concurrent users
- Provides secure API-based access
- Preserves data privacy (no third-party transmission)

The system is designed for internal enterprise deployment and controlled integration.

2. Problem Statement

Operational datasets commonly contain:

- Mixed date formats
- Misspellings and abbreviations
- Invalid numeric tokens
- Ambiguous coordinate structures
- Duplicate entries
- Missing values

Traditional Excel-based cleaning:

- Is manual
- Is inconsistent
- Has no audit trace
- Does not scale across teams
- Consumes significant operational time that could otherwise be used for analytical tasks.

This bot transforms raw Excel files into analytics-ready structured datasets while preserving traceability and adding a decision layer specifying which rows to be accepted , rejected and used conditionally post human intervention.

Further a final decision is made based on cutoff ratio of total flagged and rejected rows providing a final decision by the system whether to Accept the file or Reject it for having too many unresolvable errors or allow it Conditionally post human intervention.

3. System Architecture

3.1 Technology Stack

- FastAPI – API Layer
- PostgreSQL – Durable metadata storage (users + jobs)
- Redis – Queue management, locks, rate limits
- RQ (Redis Queue) – Background job execution
- Docker – Containerized deployment
- SQLAlchemy – ORM for DB layer

3.2 Workflow

1. User authentication (JWT-based login)
2. File upload (.xlsx , .csv)
3. Deterministic Job-ID generation (HMAC-SHA256 based on file content)
4. Atomic claim lock (prevents duplicate uploads under concurrency)
5. Background worker processes file.
6. Status endpoint returns:
 - a. queued
 - b. processing
 - c. done
 - d. failed
7. Download endpoints:
 - a. cleaned file
 - b. logs file
8. Job is deleted only after both files are downloaded

All file processing occurs within the enterprise infrastructure.

No external data transfer occurs.

3.3 Idempotency & Concurrency Control

- Job-ID derived from file content (not filename)
- Atomic Redis claim lock (SET NX EX)
- One-active-job-per-user enforcement (DB row lock)
- Atomic file publish via `os.replace()` (no partial files exposed as only the complete final file is written to the final file path)

This ensures correctness even under concurrent upload attempts.

4. Cleaning Logic (Deterministic Rules)

The bot uses rule-based deterministic transformations:

Date Handling

- Standard format: YYYY-MM-DD
- Range: 1970–2035
- Ambiguous dates flaggedged
- Invalid numeric structures rejected
- Month-first date formats (MM-DD-YYYY) are not interpreted. In ambiguous cases such as 02-03-2024, the system assumes the day-first format (DD-MM-YYYY), which aligns with common Indian date conventions
- Flagged and Rejected values returned in log file
- Date Intelligence switch has been provided for allowing flexibility regarding date cleaning strictness.
- Date tokens containing alphabetic characters within numeric positions (e.g., 2C-08-2024) are rejected instead of cleaned, as the alphabetic character may represent any digit, making deterministic correction unreliable.

- 2-letter month abbreviations and months with wrong first letters are rejected to maintain strict cleaning logic

Location Validation

- Controlled location master list (Considered Indian State and UT names, official abbreviation of state and UT names, city names – Delhi, Mumbai, Kanpur, Chandigarh). Rest of the names have been ignored to show that the bot can identify and clean based on targeted names.
- Location matching based on: $\geq 80\%$ similarity threshold
- Abbreviation matching
- Multiple locations flagged [2 locations under flagged multiple locations are displayed for convenience]; state-state, abbreviations-state considered multiple locations if present in a single cell of either of location column.
- Ambiguous names flagged
- Original value preserved and flagged & rejected values inserted in logs.

Coordinate Validation

- Enforced DMS format (latitude, longitude format)
- Requires N/S/E/W poles for clear data and prevent ambiguity.
- Minutes and seconds values exceeding 60 are corrected using carry-forward normalization.
- Out-of-bounds rejected (>90 degree for latitude, >180 degree for longitude)
- Ambiguous coordinate tokens flagged

Price Normalization

- Standard format: Rs. number
- Decimal preservation
- Lakhs/crores/Thousands converted to numeric
- Ranges of price rejected
- Negative values converted to positive

Duplicate Handling

- Duplicate rows removed post-cleaning
- Original row-id of duplicate and the row it matched to post cleaning recorded in logs

5. Outputs Generated

For each job:

1. cleaned_data.xlsx (or .csv) (cleaned file included with flags, rejected messages, row status and row usability)
2. logs_data.xlsx (or .csv) (contains full flagged rows with flag and rejected messages + row-id)

6. Performance Characteristics

Tested on:

- 4 CPU cores
- 8GB RAM
- 3 workers stable

Performance:

- ~15–16 seconds for 10,000 rows
- ~4–5 seconds for 100 rows

Actual throughput depends on:

- Worker count
- Hardware capacity
- File complexity (File on which it was tested on had errors in almost all rows)

7. Security & Access Control

- Password hashing (bcrypt mechanism)
- Authentication is implemented using signed JWT bearer tokens with configurable expiry, and all authenticated requests are expected to be transmitted only over enterprise-managed HTTPS channels.
- Ownership validation per job
- Rate limits:
 - Login endpoint (Rate limit=5 login attempts in 10 minutes, configurable)
 - Status polling (Rate limit=60 in 1 minute, configurable)
- Max file size enforced (configurable – set in environment variable file, 25MB default)
- Job timeout enforced (configurable, 10 minutes for a single processing as maximum time allotted to a worker exceeding which kills the process and marks the status as failed)
- All uploads are validated before processing.
- File paths are generated internally and not user supplied.
- Database stores information about file posting user, file details, file paths, timestamps, attempts - how many times the file was posted, Configuration version number of the program used, version hash for configuration integrity verification, all the row and duplicate row details.

No third-party API calls.

Data remains within enterprise infrastructure.

8. Deployment & Integration Model

8.1 Delivery Model

- Delivered as Docker image
- No direct access to enterprise infrastructure
- Source code sharing optional (based on agreement)

8.2 Enterprise Responsibilities

Enterprise IT will:

- Host container internally
- Provide Redis instance
- Provide PostgreSQL instance
- Configure environment variables
- Enable HTTPS via internal gateway/reverse proxy
- Deployment orchestration, including Kubernetes, container runtime configuration, networking, ingress, and internal infrastructure integration, will be handled entirely by the enterprise's IT team.

8.3 System Requirements (Recommended)

- 4 CPU cores
- 8GB RAM
- SSD recommended
- Redis persistence recommended but not mandatory as PostgreSQL maintains durable job metadata.
- The above requirements enable 3 workers in Redis to work simultaneously. Increasing the number of workers requires a proportionally higher CPU and memory resources.
- System supports horizontal scaling by increasing worker instances.

8.4 Required Services

- Redis (queue + locks)
- PostgreSQL (durable metadata storage)

9. Configuration Layer

The system supports configuration-driven rules:

- Year range configurable
- Date intelligence switch
- Version Number
- Similarity threshold configurable
- Validation toggles configurable
- Flag messages
- Location list
- Price format and pattern matching
- Job timeout
- Atomic claim key expiry time
- Job key expiry time in Redis
- Maximum file size allowed
- Rate limit attempts and time limit for it
- Thresholds and Cutoff value for file acceptability adjustable via configuration without code modification.
- All configuration changes require controlled deployment. Some configurations require changes in the environment variable file.

10. SLA & Operational Policies

- Job timeout: 10 minutes
- Failed jobs marked explicitly
- Jobs retained until both outputs downloaded
- Automatic cleanup post-download
- Duplicate uploads blocked while a job is active

11. Data Retention Policy

- Input, cleaned, and log files stored temporarily
- Deleted after both outputs downloaded
- No long-term file storage

Job metadata retained in PostgreSQL for audit.

12. Limitations

- Structured tabular data (Excel or CSV) expected
- No machine-learning-based inference (deterministic system)
- Config change requires redeployment

13. Strategic Positioning

This system is not a prompt-based AI tool.

It is a:

- Deterministic data standardization engine
- Workflow-controlled system
- Concurrency-safe processing framework
- Audit-preserving enterprise automation layer

Designed for reliability, traceability, and scalability.

SAMPLE OUTPUTS:

RAW DATA:

	A	B	C	D	E	F	G
1	location_to	location_from	Coordinate_to	Coordinate_from	price	date	
2	panem	Arunachal Pradesh	20.0°28.0'33.0"N,84.0°1.0'14.0"E	28.0°1.0'47.0"N,95.0°24.0'9.0"E		01-02-2099	
3		Gujarat	34°19'333"N,77°31'40"E	10°50'2"N,9.9323'26.0"E	Rs. 1200 to 1500	03-12-1600	
4	Bihar Gujkrat	Andhra Pradesh	15.0°24.0'29.0"N,74.0°58.0'13.0"E	15.0°54.0'2.0"N,80.0°15.0'19.0"	Rs.1.35 - 1300		
5	Uttar Pradesh BR	Ladakh	15.0°54.0'2.0"N,80.0°15.0'19.0"E	34.0°47.0'10.0",79.0°8.0'33.0"	jnvfwdn whmw	ebtbofni	
6	Odisha	Arunachal Pradesh	20.0°28.0'33.0"N,84.0°1.0'14.0"E	28.0°1.0'47.0"S,95.0°24.0'9.0"S	###680	02 03 aug 2023	
7	Br GJ	gulaRat	20.0°28.0'33.0"N,84.0°1.0'14.0"E	28.0°1.0'47.0"N,95.0°24.0'9.0"S	Rs.880 Inikdanland	1 sept 202	
8	Near Oddisha - Noid a- Branch	RNACHAL PRADESH (HYDERABAD) @	20°28'33"N 84°1'14"e	2800°1'47"n 95°24'9"e		88 02 03 akbdajbcoid 2023	
9	Punjab	Jharkhand	25°0'N , 91°54'E	28.0°1.0'47.0"N,950.0°24.0'9.0"E	rs.1.35	32-02-2026	
10	Bihar	Andhra Pradesh	15.0°24.0'29.0"N,74.0°58.0'13.0"E			1.35 june	
11	Ladakh	Ladakh	15.0°54.0'2.0"N,80.0°15.0'19.0"E	34ab,0°47.0'10.0"N,79.0°8.0'33.0"E	Rs-827.56		21031
12	Jammu	Andhra Pradesh	15.0°24.0'29.0"N,74.0°58.0'13.0"E	15.0°54.0'2.0"N,80.0°15.0'19.0"E \$	Rs. 1.35L		21031969
13	Uttar prdsh	Andhra Pradesh	15.0°24.0'29.0"N,74.0°58.0'13.0"E	"15.0°54.0'2.0N,80.015.0'19.0"E*	Rs. 1.35 l		2003
14	mbai	Andhra Pradesh	15.0°24.0'29.0"N,74.0°58.0'13.0"E	15.0°54.0'2.0"N	Rs. 1.35cr	03 Aug Sept 2023	
15	Telangana janjdanonad	bengal	20.0°28.0'33.0"N,84.0°1.0'14.0"E	95.0°24.0'9.0"E	12 Cr	04 podrick 2022	
16	Bihar Gujkrat	Arunachal	20.0°28.0'33.0"N,84.0°1.0'14.0"E	28.0°1.0'47.0"N,95.0°24.0'9.0"E -----/ abcd	13k		12032024
17	Uttar Pradesh BR	meghakya	20.0°28.0'33.0"N,84.0°1.0'14.0"E	28°1'47"N,95°24'9"E	13 k		20230312
18	bihar))))))	O))))(jammu))))))))	20.0°28.0'33.0"N,84.0°1.0'14.0"E	28.0°1.0'47.0"N,95.0°24.0'9.0"E	13 lakh	12 - Jun - 2022 ---pod	
19	miz o ram	Arunachal Pradesh	20.0°28.0'33.0"N,84.0°1.0'14.0"E	28.0°1.0'47.0"N,95.0°24.0'9.0"E	12 thousand	12/02/2023	
20	tanil ndu	chnidigarh	20.0°28.0'33.0"N,84.0°1.0'14.0"E	28.0°1.0'47.0"N,95.0°24.0'9.0"E	14 crore	2023/03/11	
21	asSaM	arunachal Pradesh	20.0°28.0'33.0"N,84.0°1.0'14.0"E	28.0°1.0'47.0"N,95.0°24.0'9.0"E	Rs.1200	2024//03---12	
22	Mumbai	Andhra Pradesh	20.0°28.0'33.0"N,84.0°1.0'14.0"E	28.0°1.0'47.0"N,95.0°24.0'9.0"E	Rs.300	2002-12-03 hbbadibaqbdbq	
23	Mummmimbai	Arunachal Pradesh	20.0°28.0'33.0"N,84.0°1.0'14.0"E	28.0°1.0'47.0"N,95.0°24.0'9.0"E	Rs.599		11-09-2001
24	Odisha	Arunachal Pradesh	20.0°28.0'33.0"N,84.0°1.0'14.0"E	28.0°1.0'47.0"N,95.0°24.0'9.0"E	Rs.300		31-05-1998
25	Odisha	Arunachal Pradesh	20.0°28.0'33.0"N,84.0°1.0'14.0"E	28.0°1.0'47.0"N,95.0°24.0'9.0"E	Rs.300		12-06-1998
26	Odisha	Arunachal Pradesh	20.0°28.0'33.0"N,84.0°1.0'14.0"E	28.0°1.0'47.0"N,95.0°24.0'9.0"E	Rs.300		31-05-1998
27	Odisha	Arunachal Pradesh	20.0°28.0'33.0"N,84.0°1.0'13.0"E	20.0°1.0'47.0"N,95.0°20.0'9.0"E	Rs.300		31-05-1998
28	Tamil Nadu	Arunachal Pradesh	20.0°27.0'33.0"N,85.0°1.0'14.0"E	28.0°1.0'47.0"N,95.0°24.0'9.0"E	Rs.300		31-05-1998
29	Telangana	Andhra Pradesh	21.0°28.0'33.0"N,84.0°1.0'14.0"E	29.0°1.0'47.0"N,95.0°24.0'9.0"E	Rs.300		31-05-1998
30	Odisha	Arunachal Pradesh	20.0°28.0'33.0"N,84.0°1.0'14.0"E	28.0°1.0'47.0"N,95.0°24.0'9.0"E	Rs.300	31-may 1998	
31							

CLEANED FILE PRODUCED:

A	B	C	D	E
row-id	location_to	location_from	Coordinate_to	Coordinate_from
1	[REJECTED] Could not find match location: panem, Similarity score:60.0%	Arunachal Pradesh	20.0°28.0'33.0"N,84.0°1.0'14.0"E	28.0°1.0'47.0"N,95.0°24.0'9.0"E
2	[REJECTED] No Location	Gujarat	34.0°24.0'33.0"N,77.0°31.0'40.0"E	[REJECTED] Missing coordinates: 10°59'2"N,9,9323°26.0"E
3	[FLAG] Bihar GujArat Row: 3, Multiple locations found: Gujarat (100%), Bihar (100%)	Andhra Pradesh	15.0°24.0'29.0"N,74.0°58.0'13.0"E	[REJECTED] Invalid coordinates : Multiple or no E or W found: 15.0°54.0'2.0"N,80.0°15.0'19.0"
4	[FLAG] Abbreviation - State conflict: Uttar Pradesh BR -> BR, Uttar Pradesh	Ladakh	15.0°54.0'2.0"N,80.0°15.0'19.0"E	[REJECTED] N/S/E/W not found : 34.0°47.0'10.0",79.0°8.0'33.0"
5	Odisha	Arunachal Pradesh	20.0°28.0'33.0"N,84.0°1.0'14.0"E	[REJECTED] Invalid coordinates : Multiple or no E or W found: 28.0°1.0'47.0"S,95.0°24.0'9.0"S
6	[FLAG] Multiple locations found: Br GJ -> BR, GJ	Gujarat	20.0°28.0'33.0"N,84.0°1.0'14.0"E	[REJECTED] Invalid coordinates : Multiple or no N or S found: 28.0°1.0'47.0"N,95.0°24.0'9.0"S
7	Odisha	Arunachal Pradesh	20.0°28.0'33.0"N,84.0°1.0'14.0"E	[REJECTED] Out of Bound Latitude: 2800°1'47"N 95°24'9"E
8	Punjab	Jharkhand	25.0°0.0'N,91.0°54.0"E	[REJECTED] Out of Bound Longitude: 28.0°1.0'47.0"N,950.0°24.0'9.0"E
9	Bihar	Andhra Pradesh	15.0°24.0'29.0"N,74.0°58.0'13.0"E	[REJECTED] Missing coordinates: NAN
10	Ladakh	Ladakh	15.0°54.0'2.0"N,80.0°15.0'19.0"E	[FLAG] Ambiguous coordinates words between digits: 34A8.0°47.0'10.0"N,79.0°8.0'33.0"E
11	Jammu and Kashmir	Andhra Pradesh	15.0°24.0'29.0"N,74.0°58.0'13.0"E	[REJECTED] Invalid coordinates : Multiple or no N or S found: 15.0°54.0'2.0"N,80.0°15.0'19.0"E
12	Uttar Pradesh	Andhra Pradesh	15.0°24.0'29.0"N,74.0°58.0'13.0"E	[REJECTED] Missing coordinates: "15.0°54.0'2.0N,80.015.0'19.0"E"
13	Mumbai	Andhra Pradesh	15.0°24.0'29.0"N,74.0°58.0'13.0"E	[REJECTED] Longitude is missing: 15.0°54.0'2.0"N
14	Telangana	West Bengal	20.0°28.0'33.0"N,84.0°1.0'14.0"E	[REJECTED] Latitude is missing: 95.0°24.0'9.0"E
15	[FLAG] Bihar GujArat Row: 15, Multiple locations found: Gujarat (100%), Bihar (100%)	Arunachal Pradesh	20.0°28.0'33.0"N,84.0°1.0'14.0"E	28.0°1.0'47.0"N,95.0°24.0'9.0"E
16	[FLAG] Abbreviation - State conflict: Uttar Pradesh BR -> BR, Uttar Pradesh	Meghalaya	20.0°28.0'33.0"N,84.0°1.0'14.0"E	28.0°1.0'47.0"N,95.0°24.0'9.0"E
17	Bihar	Jammu and Kashmir	20.0°28.0'33.0"N,84.0°1.0'14.0"E	28.0°1.0'47.0"N,95.0°24.0'9.0"E
18	Mizoram	Arunachal Pradesh	20.0°28.0'33.0"N,84.0°1.0'14.0"E	28.0°1.0'47.0"N,95.0°24.0'9.0"E
19	Tamil Nadu	Chandigarh	20.0°28.0'33.0"N,84.0°1.0'14.0"E	28.0°1.0'47.0"N,95.0°24.0'9.0"E
20	Assam	Arunachal Pradesh	20.0°28.0'33.0"N,84.0°1.0'14.0"E	28.0°1.0'47.0"N,95.0°24.0'9.0"E
21	Mumbai	Andhra Pradesh	20.0°28.0'33.0"N,84.0°1.0'14.0"E	28.0°1.0'47.0"N,95.0°24.0'9.0"E
22	Mumbai	Arunachal Pradesh	20.0°28.0'33.0"N,84.0°1.0'14.0"E	28.0°1.0'47.0"N,95.0°24.0'9.0"E
23	Odisha	Arunachal Pradesh	20.0°28.0'33.0"N,84.0°1.0'14.0"E	28.0°1.0'47.0"N,95.0°24.0'9.0"E
24	Odisha	Arunachal Pradesh	20.0°28.0'33.0"N,84.0°1.0'14.0"E	28.0°1.0'47.0"N,95.0°24.0'9.0"E
25	Odisha	Arunachal Pradesh	20.0°28.0'33.0"N,84.0°1.0'13.0"E	20.0°1.0'47.0"N,95.0°20.0'9.0"E
26	Tamil Nadu	Arunachal Pradesh	20.0°27.0'33.0"N,85.0°1.0'14.0"E	28.0°1.0'47.0"N,95.0°24.0'9.0"E
27	Telangana	Andhra Pradesh	21.0°28.0'33.0"N,84.0°1.0'14.0"E	29.0°1.0'47.0"N,95.0°24.0'9.0"E

F	G	H	I
price	date	Row_Status	Usable
[REJECTED] No Price	[REJECTED] Year out of allowed range: 2099-02-01 00:00:00	REJECTED	NO
[REJECTED] Range of price prohibited: Rs. 1200 to 1500	[REJECTED] INVALID DATE: 03-12-1600	REJECTED	NO
[REJECTED] Range of price prohibited: Rs.1.35 - 1300	[REJECTED] No Date	REJECTED	NO
[REJECTED] No numeric value to recover -> requires human input: jnvljwn vlvr	[REJECTED] INVALID DATE couldn't parse: ebfbofnl	REJECTED	NO
Rs.880	[FLAG] Multiple months found 02 03 aug 2023	REJECTED	NO
Rs.880	[REJECTED] Invalid year 1 sept 202	REJECTED	NO
Rs.88	2023-03-02	REJECTED	NO
Rs.1.35	[REJECTED] INVALID DATE: 32-02-2026	REJECTED	NO
Rs.1.35	[REJECTED] INVALID DATE couldn't parse: june	REJECTED	NO
Rs.827.56	[REJECTED] INVALID DATE couldn't parse: 21031	REJECTED	NO
Rs.135000	1989-03-21	REJECTED	NO
Rs.135000	[REJECTED] INVALID DATE couldn't parse: 2003	REJECTED	NO
Rs.13500000	[FLAG] Multiple months found 03 Aug Sept 2023	REJECTED	NO
Rs.120000000	[FLAG] No month found 04 podrick 2022	REJECTED	NO
Rs.13000	2024-03-12	FLAGGED	CONDITIONAL
Rs.13000	2023-03-12	FLAGGED	CONDITIONAL
Rs.1300000	2022-06-12	OK	YES
Rs.12000	2023-02-12	OK	YES
Rs.140000000	2023-03-11	OK	YES
Rs.1200	2024-03-12	OK	YES
Rs.300	2002-12-03	OK	YES
Rs.599	2001-09-11	OK	YES
Rs.300	1998-05-31	OK	YES
Rs.300	1998-06-12	OK	YES
Rs.300	1998-05-31	OK	YES
Rs.300	1998-05-31	OK	YES
Rs.300	1998-05-31	OK	YES

LOGS FILE PRODUCED:

A	B	C	D	E
ow_id	location_to	location_from	Coordinate_to	Coordinate_from
1	[REJECTED] Could not find match location: panem, Similarity score:60.0%	Arunachal Pradesh	20.0°28.0'33.0"N,84.0°1.0'14.0"E	28.0°1.0'47.0"N,95.0°24.0'9.0"E
2	[REJECTED] No Location	Gujarat	34.0°24.0'33.0"N,77.0°31.0'40.0"E	[REJECTED] Missing coordinates: 10°59'2 N°9, 9323°26.0"E
3	[FLAG] Bihar GujArat Row: 3, Multiple locations found: Gujarat (100%), Bihar (100%)	Andhra Pradesh	15.0°24.0'29.0"N,74.0°58.0'13.0"E	[REJECTED] Invalid coordinates : Multiple or no E or W found: 15.0°54.0'2.0"N,80.0°15.0'19.0"
4	[FLAG] Abbreviation - State conflict: Uttar Pradesh BR -> BR, Uttar Pradesh	Ladakh	15.0°54.0'2.0"N,80.0°15.0'19.0"E	[REJECTED] N/S/E/W not found : 34.0°47.0'10.0",79.0°8.0'33.0"
5	Odisha	Arunachal Pradesh	20.0°28.0'33.0"N,84.0°1.0'14.0"E	[REJECTED] Invalid coordinates : Multiple or no E or W found: 28.0°1.0'47.0"S,95.0°24.0'9.0"S
6	[FLAG] Multiple locations found: Br GJ -> BR, GJ	Gujarat	20.0°28.0'33.0"N,84.0°1.0'14.0"E	[REJECTED] Invalid coordinates : Multiple or no N or S found: 28.0°1.0'47.0"N,95.0°24.0'9.0"S
7	Odisha	Arunachal Pradesh	20.0°28.0'33.0"N,84.0°1.0'14.0"E	[REJECTED] Out of Bound Latitude: 2800°1'47"N 95°24'9"E
8	Punjab	Jharkhand	25.0°0.0'N,91.0°54.0'E	[REJECTED] Out of Bound Longitude: 28.0°1.0'47.0"N,950.0°24.0'9.0"E
9	Bihar	Andhra Pradesh	15.0°24.0'29.0"N,74.0°58.0'13.0"E	[REJECTED] Missing coordinates: NAN
10	Ladakh	Ladakh	15.0°54.0'2.0"N,80.0°15.0'19.0"E	[FLAG] Ambiguous coordinates words between digits: 34AB.0°47.0'10.0"N,79.0°8.0'33.0"E
11	Jammu and Kashmir	Andhra Pradesh	15.0°24.0'29.0"N,74.0°58.0'13.0"E	[REJECTED] Invalid coordinates : Multiple or no N or S found: 15.0°54.0'2.0"N,80.0°15.0'19.0"E
12	Uttar Pradesh	Andhra Pradesh	15.0°24.0'29.0"N,74.0°58.0'13.0"E	[REJECTED] Missing coordinates: "15.0°54.0'2.ON,80.015.0'19.0"E"
13	Mumbai	Andhra Pradesh	15.0°24.0'29.0"N,74.0°58.0'13.0"E	[REJECTED] Longitude is missing: 15.0°54.0'2.0"N
14	Telangana	West Bengal	20.0°28.0'33.0"N,84.0°1.0'14.0"E	[REJECTED] Latitude is missing: 95.0°24.0'9.0"E
15	[FLAG] Bihar GujArat Row: 15, Multiple locations found: Gujarat (100%), Bihar (100%)	Arunachal Pradesh	20.0°28.0'33.0"N,84.0°1.0'14.0"E	28.0°1.0'47.0"N,95.0°24.0'9.0"E
16	[FLAG] Abbreviation - State conflict: Uttar Pradesh BR -> BR, Uttar Pradesh	Meghalaya	20.0°28.0'33.0"N,84.0°1.0'14.0"E	28.0°1.0'47.0"N,95.0°24.0'9.0"E

	F	G	H	I
1	price	date	Duplicate Rows	
2	[REJECTED]-No Price	[REJECTED] Year out of allowed range: 2099-02-01 00:00:00	Duplicate rows: {25, 29}. Matched with rows: [23, 25, 29],[23, 25, 29]	
3	[REJECTED] Range of price prohibited: Rs. 1	[REJECTED] INVALID DATE: 03-12-1600	Duplicate rows: {25, 29}. Matched with rows: [23, 25, 29],[23, 25, 29]	
4	[REJECTED] Range of price prohibited: Rs.1.	[REJECTED] No Date	Duplicate rows: {25, 29}. Matched with rows: [23, 25, 29],[23, 25, 29]	
5	[REJECTED] No numeric value to recover ->	[REJECTED] INVALID DATE couldn't parse: ebfbofnl	Duplicate rows: {25, 29}. Matched with rows: [23, 25, 29],[23, 25, 29]	
6	Rs.880	[FLAG] Multiple months found 02 03 aug 2023	Duplicate rows: {25, 29}. Matched with rows: [23, 25, 29],[23, 25, 29]	
7	Rs.880	[REJECTED] Invalid year 1 sept 202	Duplicate rows: {25, 29}. Matched with rows: [23, 25, 29],[23, 25, 29]	
8	Rs.88	2023-03-02	Duplicate rows: {25, 29}. Matched with rows: [23, 25, 29],[23, 25, 29]	
9	Rs.1.35	[REJECTED] INVALID DATE: 32-02-2026	Duplicate rows: {25, 29}. Matched with rows: [23, 25, 29],[23, 25, 29]	
10	Rs.1.35	[REJECTED] INVALID DATE couldn't parse: june	Duplicate rows: {25, 29}. Matched with rows: [23, 25, 29],[23, 25, 29]	
11	Rs.827.56	[REJECTED] INVALID DATE couldn't parse: 21031	Duplicate rows: {25, 29}. Matched with rows: [23, 25, 29],[23, 25, 29]	
12	Rs.135000	1989-03-21	Duplicate rows: {25, 29}. Matched with rows: [23, 25, 29],[23, 25, 29]	
13	Rs.135000	[REJECTED] INVALID DATE couldn't parse: 2003	Duplicate rows: {25, 29}. Matched with rows: [23, 25, 29],[23, 25, 29]	
14	Rs.13500000	[FLAG] Multiple months found 03 Aug Sept 2023	Duplicate rows: {25, 29}. Matched with rows: [23, 25, 29],[23, 25, 29]	
15	Rs.120000000	[FLAG] No month found 04 podrick 2022	Duplicate rows: {25, 29}. Matched with rows: [23, 25, 29],[23, 25, 29]	
16	Rs.13000	2024-03-12	Duplicate rows: {25, 29}. Matched with rows: [23, 25, 29],[23, 25, 29]	
17	Rs.13000	2023-03-12	Duplicate rows: {25, 29}. Matched with rows: [23, 25, 29],[23, 25, 29]	
18				

Duplicate rows list both the duplicate record and the original row it matches. Row-id+1=Excel Index [In original file]

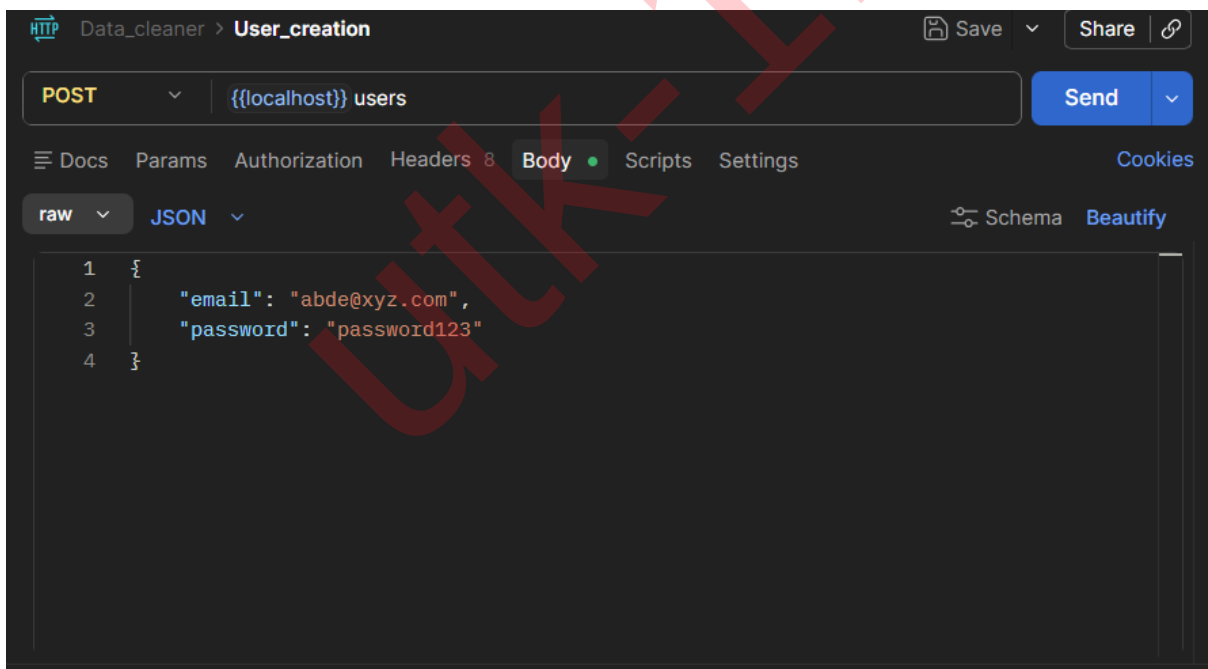
**SAMPLE LOGS OUTPUT FOR DATE WHEN DATE INTELLIGENCE SWITCH IS
TURNED OFF (FOR SAME SAMPLE ROWS OF INPUT PROVIDED ABOVE):**

G	H
date	Duplicate Rows
[REJECTED] Year out of allowed range: 2099-02-01 00:00:00	Duplicate rows: {25}. Matched with rows: [23, 25]
[REJECTED] INVALID DATE couldn't parse: 03-12-1600	Duplicate rows: {25}. Matched with rows: [23, 25]
[REJECTED] No Date	Duplicate rows: {25}. Matched with rows: [23, 25]
[REJECTED] INVALID DATE: ebfbofnl	Duplicate rows: {25}. Matched with rows: [23, 25]
[REJECTED] INVALID DATE couldn't parse: 02 03 aug 2023	Duplicate rows: {25}. Matched with rows: [23, 25]
[REJECTED] INVALID DATE: 1 sept 202	Duplicate rows: {25}. Matched with rows: [23, 25]
[REJECTED] INVALID DATE couldn't parse: 02 03 akbdajbcojd 2023	Duplicate rows: {25}. Matched with rows: [23, 25]
[REJECTED] INVALID DATE couldn't parse: 32-02-2026	Duplicate rows: {25}. Matched with rows: [23, 25]
[REJECTED] INVALID DATE: june	Duplicate rows: {25}. Matched with rows: [23, 25]
[REJECTED] INVALID DATE: 21031	Duplicate rows: {25}. Matched with rows: [23, 25]
1989-03-21	Duplicate rows: {25}. Matched with rows: [23, 25]
[REJECTED] INVALID DATE: 2003	Duplicate rows: {25}. Matched with rows: [23, 25]
[REJECTED] INVALID DATE: 03 Aug Sept 2023	Duplicate rows: {25}. Matched with rows: [23, 25]
[REJECTED] INVALID DATE: 04 podrick 2022	Duplicate rows: {25}. Matched with rows: [23, 25]
2024-03-12	Duplicate rows: {25}. Matched with rows: [23, 25]
2023-03-12	Duplicate rows: {25}. Matched with rows: [23, 25]
[REJECTED] INVALID DATE: 12 - Jun - 2022 ----pod	Duplicate rows: {25}. Matched with rows: [23, 25]
[REJECTED] INVALID DATE couldn't parse: 2002-12-03 hbbadibaqqbdq	Duplicate rows: {25}. Matched with rows: [23, 25]
[REJECTED] INVALID DATE: 31-may 1998	Duplicate rows: {25}. Matched with rows: [23, 25]

Month abbreviations were not cleaned rather flagged, any format of date
which is broken and / or has any extra words or noise has not been cleaned
rather flagged for targeted human intervention

Following images of Postman are shared to highlight the mechanism of accessing the data cleaner:

USER CREATION:



USER LOGIN WITH CORRECT PASSWORD

The screenshot shows a REST client interface for a POST request to `{{local}} login`. The request body is set to `form-data` with two fields: `username` (value: `abde@xyz.com`) and `password` (value: `password123`). The response is a 200 OK status with a response time of 461 ms and a body size of 288 B. The response body is shown in JSON format, containing an `access_token` (redacted with a black box and labeled "YOUR TOKEN") and a `token_type` of `"bearer"`.

HTTP Data_cleaner > User_login Save Share

POST {{local}} login Send

Docs Params Authorization Headers 8 Body Scripts Settings Cookies

form-data

Key	Value	Description	Bulk Edit	...
✓ username	Text abde@xyz.com			
✓ password	Text password123			
Key	Text Value	Description		

Body 200 OK • 461 ms • 288 B Save Response

{ } JSON Preview Visualize

```
1 {
2   "access_token": "YOUR TOKEN"
3   "token_type": "bearer"
4 }
```

USER LOGIN WITH WRONG PASSWORD

HTTP Data_cleaner > User_login Save Share

POST {{local}} login Send

Docs Params Authorization Headers 8 Body Scripts Settings Cookies

☐ none ☒ form-data ☐ x-www-form-urlencoded ☐ raw ☐ binary ☐ GraphQL

Key		Value	Description	Bulk Edit	...
✓ username	Text	abde@xyz.com			
✓ password	Text	password1111111			
Key	Text	Value	Description		

Body Cookies Headers 4 Test Results 403 Forbidden • 462 ms • 164 B • Save Response ...

{ } JSON Preview Try with different auth credentials

```
1 {
2   "detail": "Invalid Credentials"
3 }
```

POSTING FILE

The screenshot shows a REST client interface with the following details:

- URL:** `{{local}} posts`
- Method:** `POST`
- Body Type:** `form-data`
- Form Data:**

Key	Value	Description
☒ file	Dummy_test.xlsx	
- Response:**
 - Status:** 202 Accepted
 - Time:** 59 ms
 - Size:** 209 B
 - Content Type:** JSON
 - Body:**

```
{  "filename": "Dummy_test.xlsx",  "job_id": "2df95e44-c84d-549f-a25c-4a012d93db76"}
```

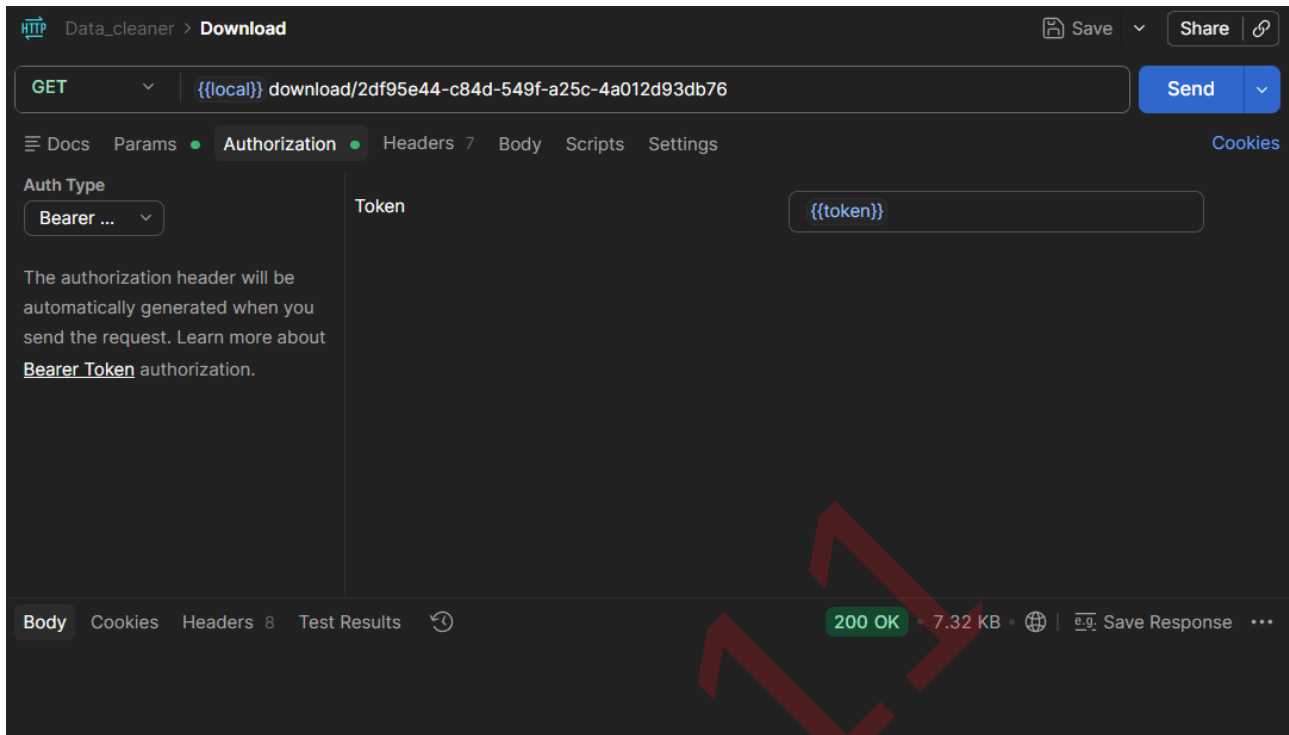
VIEWING STATUS

The screenshot displays a REST client interface with the following components:

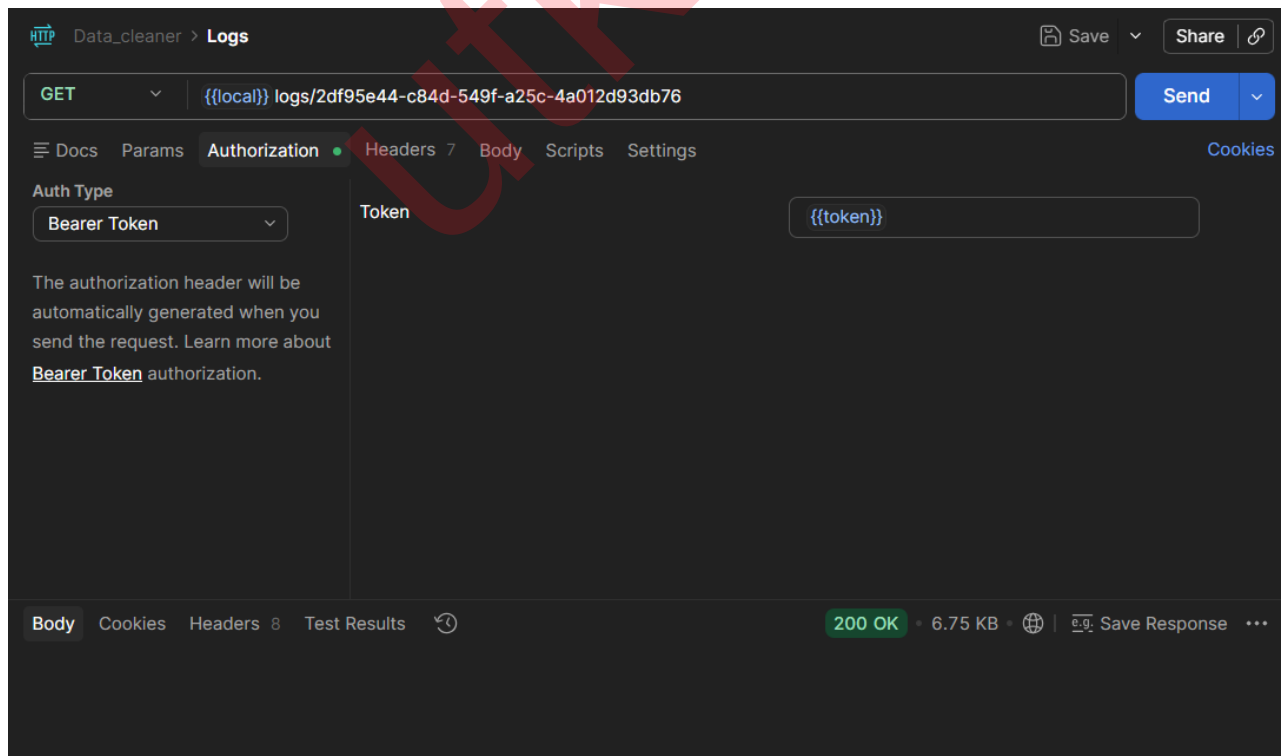
- Top Bar:** Shows the HTTP method `GET` and the URL `{{local}} status/2df95e44-c84d-549f-a25c-4a012d93db76`. Buttons for `Save`, `Share`, and `Send` are present.
- Auth Section:** The `Authorization` tab is active, showing `Bearer ...` as the auth type and `{{token}}` as the token value. A note explains that the authorization header will be automatically generated.
- Response Section:** The `Body` tab is active, showing a `200 OK` status with a response time of `22 ms` and a size of `610 B`. The response is in `JSON` format.
- JSON Response:** The response body is a JSON object with the following structure:

```
1 {
2   "job_id": "2df95e44-c84d-549f-a25c-4a012d93db76",
3   "status": "done",
4   "created_at": "2026-04-20T19:39:06.938651+05:30",
5   "updated_at": "2026-04-20T19:39:09.226409+05:30",
6   "finished_at": "2026-04-20T19:39:09.226409+05:30",
7   "total_rows": 29,
8   "total_columns": 6,
9   "clean_rows": 5,
10  "corrected_rows": 6,
11  "rejected_rows": 14,
12  "flagged_rows": 2,
13  "duplicate_rows": 2,
14  "decision": "REJECTED",
15  "download_cleaned": "/download/2df95e44-c84d-549f-a25c-4a012d93db76",
16  "download_logs": "/logs/2df95e44-c84d-549f-a25c-4a012d93db76"
17 }
```

DOWNLOADING CLEANED FILE



DOWNLOADING LOGS FILE



USER DATABASE TABLE

Data Output Messages Notifications				
Showing rows: 1 to 2				
	id [PK] integer	email character varying	password character varying	Time timestamp with time zone
1	1	abde@xyz.com	\$2b\$12\$7BWEV8j3jq4qAoqe2vkviOxdzVPKebesmwJU80aNTITX2r6a...	2026-04-05 20:51:43.271433+05:...

AUDIT JOB DATABASE TABLE

Data Output Messages Notifications						
Showing rows: 1 to 10						
	job_id [PK] character varying	user_email character varying	filename character varying	file_size_bytes bigint	mime_type character varying	final_status character varying
1	2df95e44-c84d-549f-a25c-4a012d93db...	abde@xyz.com	Dummy_test.xlsx	11969	application/vnd.openxmlformats-officedocument.spreadsheetml.s...	done
2	bdaedba6-2965-513e-b414-7e92836a0...	abde@xyz.com	Dummy_test.xlsx	11812	application/vnd.openxmlformats-officedocument.spreadsheetml.s...	done
3	1e3a7d33-ce25-588d-be14-608399489...	abde@xyz.com	Dummy_test.xlsx	11832	application/vnd.openxmlformats-officedocument.spreadsheetml.s...	done
4	2b8d19e2-9046-52e6-8e0b-9df92f5d99...	abde@xyz.com	Dummy_test.xlsx	11847	application/vnd.openxmlformats-officedocument.spreadsheetml.s...	done
5	8f7662c9-9310-5f25-becc-09ae4898efe9	abde@xyz.com	Dummy_test.xlsx	9790	application/vnd.openxmlformats-officedocument.spreadsheetml.s...	done
6	05bb7c29-d46f-562b-a711-cf07adb87d...	abde@xyz.com	Dummy_test.xlsx	9594	application/vnd.openxmlformats-officedocument.spreadsheetml.s...	done
7	c185956c-766e-5cca-9a4e-76587b640f...	abde@xyz.com	Dummy_test.xlsx	9380	application/vnd.openxmlformats-officedocument.spreadsheetml.s...	done
8	57659560-e930-5593-b76f-ff570fb6d5d9	abde@xyz.com	Dummy_test.xlsx	9613	application/vnd.openxmlformats-officedocument.spreadsheetml.s...	done

Data Output Messages Notifications							
Showing rows: 1 to 10							
	downloaded_cleaned boolean	downloaded_logs boolean	created_at timestamp with time zone	finished_at timestamp with time zone	updated_at timestamp with time zone	attempts integer	data_intelligence_switch boolean
1	true	true	2026-04-20 19:39:06.938651+05:...	2026-04-20 19:39:09.226409+05:...	2026-04-20 19:39:53.46445+05:30	1	true
2	true	true	2026-04-15 19:47:15.207681+05:...	2026-04-15 19:47:17.141464+05:...	2026-04-15 19:48:08.635135+05:...	1	true
3	true	true	2026-04-15 18:00:01.330481+05:...	2026-04-15 18:00:04.708254+05:...	2026-04-15 18:01:02.016624+05:...	1	true
4	true	true	2026-04-15 16:51:55.073723+05:...	2026-04-15 16:51:57.578841+05:...	2026-04-15 16:53:08.223931+05:...	1	true
5	true	true	2026-04-13 21:37:38.233804+05:...	2026-04-14 15:28:24.054528+05:...	2026-04-14 15:28:39.796873+05:...	4	true
6	true	true	2026-04-13 20:45:10.212218+05:...	2026-04-13 21:04:15.502892+05:...	2026-04-13 21:04:32.343275+05:...	2	true
7	true	true	2026-04-13 17:16:58.02315+05:30	2026-04-13 17:54:37.027795+05:...	2026-04-13 17:55:06.648745+05:...	4	true
8	true	true	2026-04-13 17:40:29.428179+05:...	2026-04-13 17:40:29.757661+05:...	2026-04-13 17:41:34.213066+05:...	1	true

Data Output Messages Notifications			
Showing rows: 1 to 10			
	input_path character varying	output_path character varying	log_path character varying
1		(Temp\upload_jvj47_24.x...	(Temp\cleaned_l99yqSow.x...
2		(Temp\upload_iswp6t7s...	(Temp\cleaned_xox1ztuq.xl...
3		(Temp\upload_ac24vtk1...	(Temp\cleaned_qrtc705a.xl...
4		(Temp\upload_v6q20lj.xl...	(Temp\cleaned_zjsm898b...
5		(Temp\upload_7053j5ku...	(Temp\cleaned_y0s26n1s.x...
6		(Temp\upload_jwfck7jf.xl...	(Temp\cleaned_jhvjhsk.xlsx
7		(Temp\upload_bsuxchpx...	(Temp\cleaned_seo883tf.xl...
8		(Temp\upload_kfx22t0c...	(Temp\cleaned_olutiaeg.xlsx

Data Output Messages Notifications											
Showing rows: 1 to 10											
	version character varying	version_hash character varying	total_rows integer	total_columns integer	clean_rows integer	corrected_rows integer	rejected_rows integer	flagged_rows integer	decision character varying	duplicate_rows integer	
1	3yw...	1.0.0	c033bd8de9b0d99...	29	6	5	6	14	2	REJECTED	2
2	req1...	1.0.0	c033bd8de9b0d99...	28	6	3	5	14	2	REJECTED	4
3	qd.x...	1.0.0	c033bd8de9b0d99...	28	6	3	5	14	2	REJECTED	4
4	lnwq...	1.0.0	0b5326ace691c8a...	28	6	3	7	14	0	REJECTED	4
5	zfo...	1.0.0	0b5326ace691c8a...	9	6	3	2	1	2	REJECTED	1
6	jo.v...	1.0.0	0b5326ace691c8a...	7	6	0	0	6	0	REJECTED	1
7	vin...	1.0.0	0b5326ace691c8a...	4	6	3	0	1	0	REJECTED	0
8	hbr...	1.0.0	0b5326ace691c8a...	7	6	0	1	5	0	REJECTED	1

Conclusion

The Intelligent Data Cleaning Bot enables:

- Workforce time reduction
- Scalable concurrent processing
- Structured audit trace
- Deterministic data standardization
- Secure internal deployment

It transforms raw datasets into analytics-ready structured outputs while maintaining human oversight for flagged and rejected cases. It has been designed with the view of keeping Human in the loop concept for final checking and targeted intervention. The system performs most of the deterministic cleaning tasks automatically, allowing human reviewers to focus only on flagged edge cases and provides insight into the data regarding file errors and decision regarding the file, thus reducing manual effort significantly, allowing the employees to handle greater number of files per day, save more time and send larger amount of data within lesser amount of time for exploratory analysis enhancing robust and faster decision making.